LESSON 4 • TERM 4 • LEVEL 1

Cubes and Views

The same thing looks different from the top

TODAY WE WILL

Count every cube, even one we cannot see.

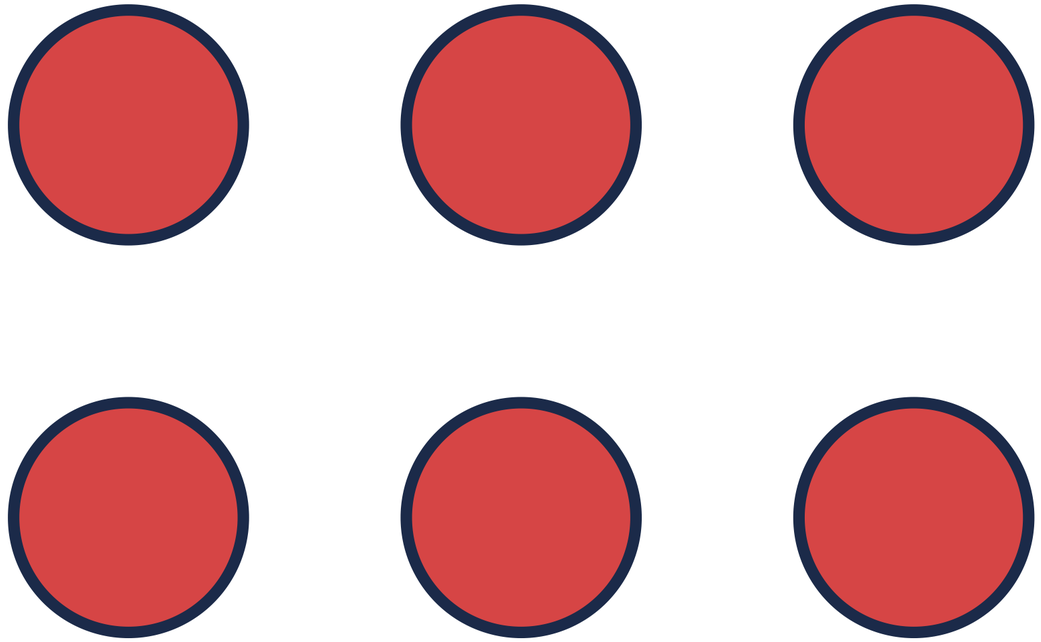
Look at a stack from the top.

Fold a paper and find the holes.



Warm-up: rows

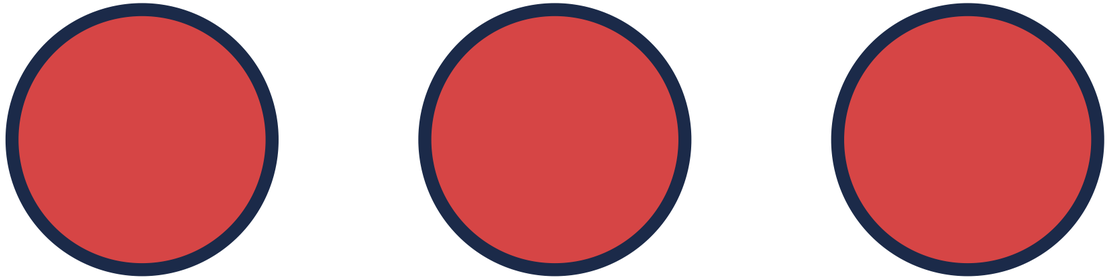
2–8 MIN



Count the counters by rows. How many rows? How many in each row? How many in all?

Doubles

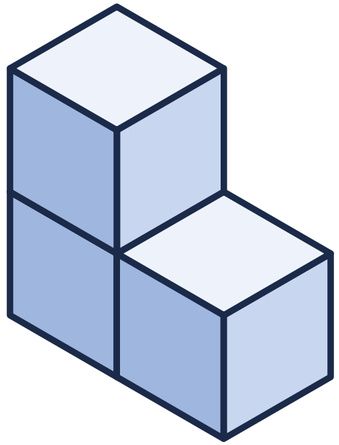
2–8 MIN



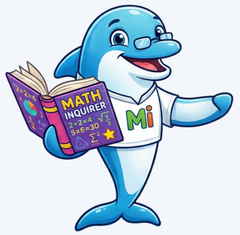
A double is the same number twice. Double 3 is 3 and 3 again. Double 5?

Count the cubes

2–8 MIN



Touch each cube. How many cubes are there?



Take your time!

Sneak-peek puzzles

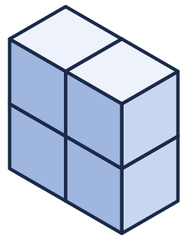
8–12 MIN

Ideas we have not learnt yet. A guess is a great start. Nobody is marked.

PUZZLE 1: anyone can try

How many cubes are in the picture?

Write the number.

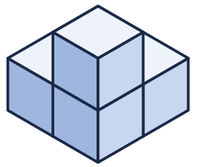


PUZZLE 2: think

The stack has 5 cubes.

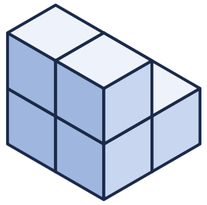
The picture shows only 4.

Where is the 5th cube? I know because ___.



Explore: build the stack

12–23 MIN

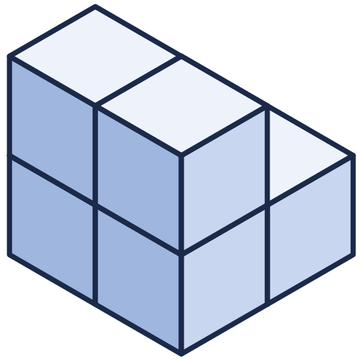


1	1
2	2

The numbers say how many blocks are in each stack. Build the stack. How many blocks do you use?

6 blocks, 1 hidden

12–23 MIN



The stack has 6 blocks. The picture shows only 5. One block is hidden behind the front tower.

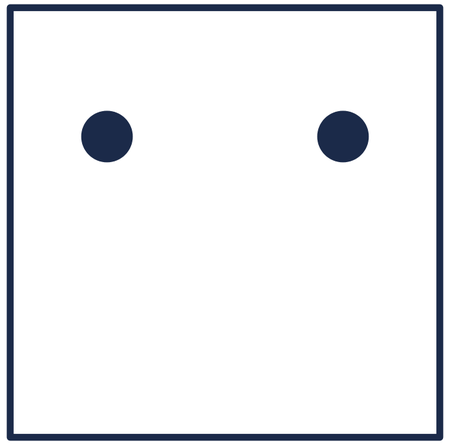
Explore: fold and open

12–23 MIN

1. Fold your paper square in half. Make the edges meet.
2. Make one pencil dot on the folded paper, away from the fold.
3. How many dots will there be when you open it?

Open it up

12–23 MIN



One dot on the folded paper. Two dots when it is open.

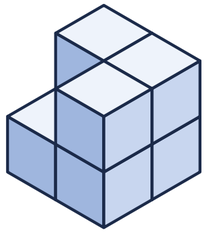
What did we notice?

12–23 MIN

- What was hidden when you looked at the picture of the stack?
- Where were the two dots after you opened the paper?

Count every cube

23–34 MIN

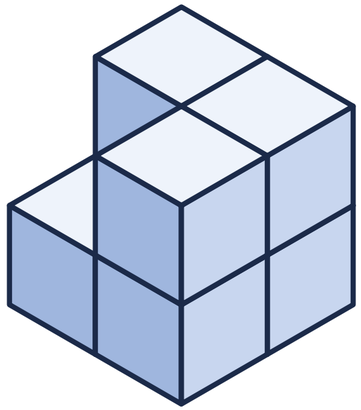


2	2
1	2

The top view says how many cubes are in each stack. How many cubes can you see in the picture? How many are there in all?

7 cubes in all

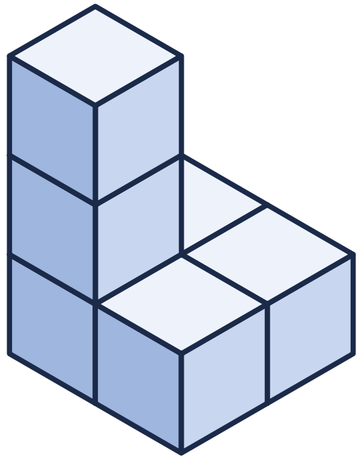
23–34 MIN



We can see 6 cubes. There are 7 in all: 2 + 2 + 1 + 2. One cube is hidden.

The top view as numbers

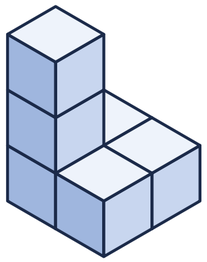
23–34 MIN



Look from the top. Write a number for each stack. Draw 4 squares like a window.

The top view

23–34 MIN

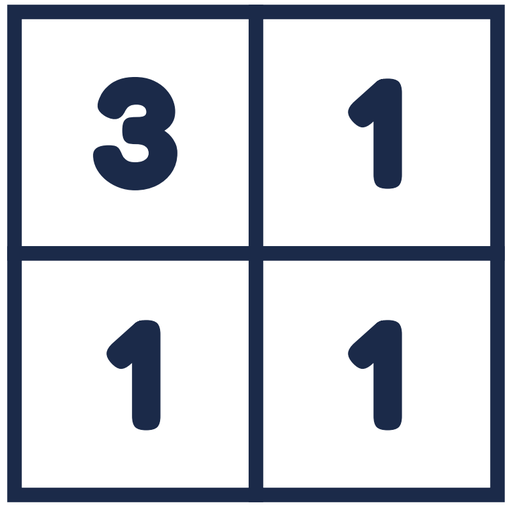


1	1
3	1

Top row: 1 and 1. Bottom row: 3 and 1. There are 6 cubes in all.

Build from the top view

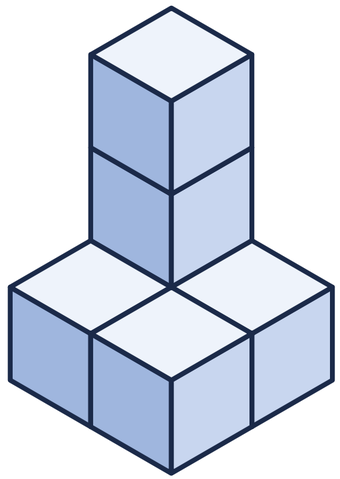
23–34 MIN



Build the stack with your blocks. How many cubes are there in all?

6 cubes

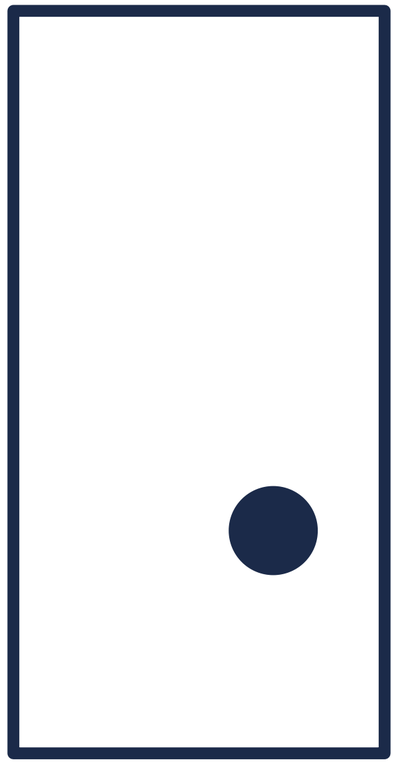
23–34 MIN



3 + 1 + 1 + 1 = 6 cubes. The tall stack is at the back on the left.

Fold once, punch once

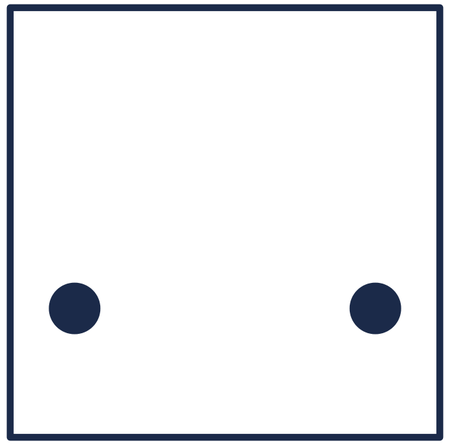
23–34 MIN



A paper is folded in half. The fold is on the left. One hole is punched. How many holes when it is opened?

Two holes

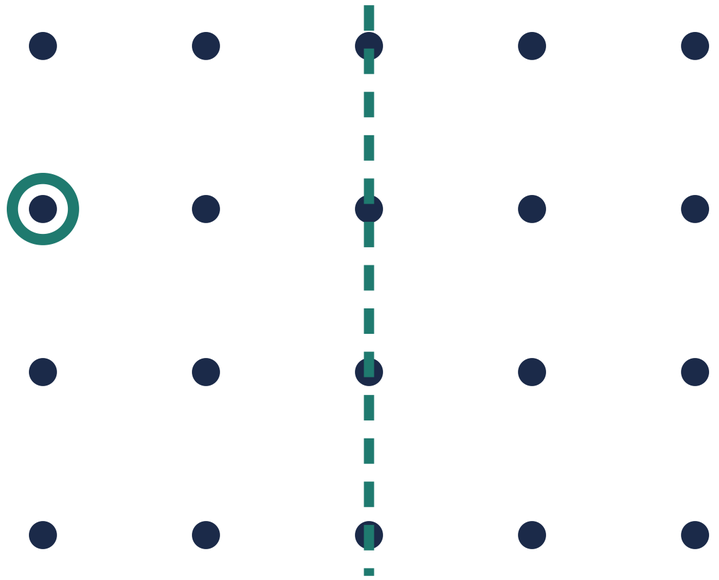
23–34 MIN



2 holes. The fold is in the middle. The holes are the same distance from it.

Same distance from the fold

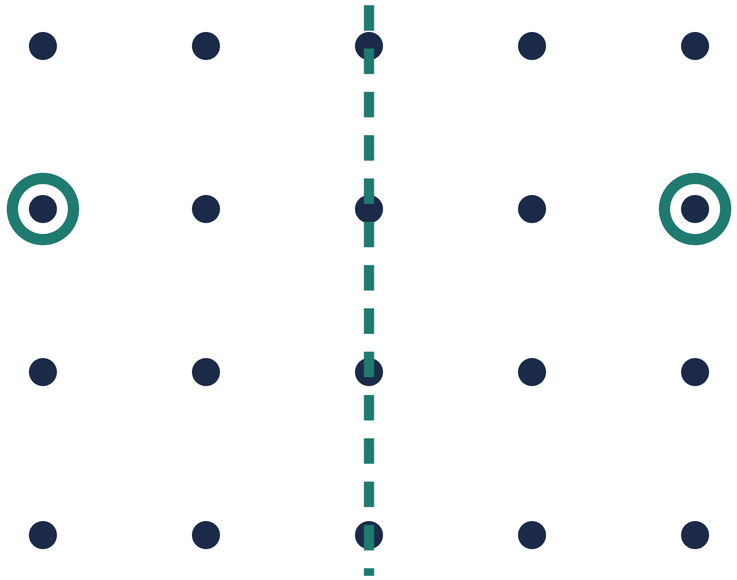
23–34 MIN



The paper is folded along the dashed line. One hole is ringed. Where is the other hole?

2 steps each side

23–34 MIN



The other hole is 2 steps from the dashed line, on the other side.

34–38 MIN

Fold and unfold break

Stand up. Fold your arms in half. Unfold them.

Make a tall stack with your arms. Make a flat one.

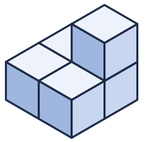
Sip some water.

Workbook page 1: Start here

38–70 MIN

Questions 1 to 3. Take your time.

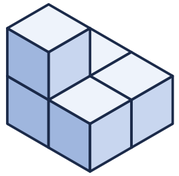
S1 Look at the stack of cubes.



S2 The stack has 4 cubes in all.



S3 Which top view matches this stack?



2	1
1	1

1	1
2	1

1	2
1	1

Check page 1

38–70 MIN

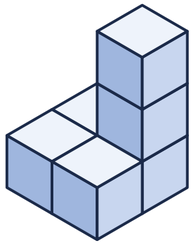
S1	5 cubes
S2	3; 1
S3	B

Workbook page 2: Start here

38–70 MIN

Questions 4 to 6. Take your time.

S4 Look at the stack. Write how many cubes are in each stack on the top view.





S5 A paper square is folded so that the top edge meets the bottom edge.



S6 The top view shows how many cubes are in each stack.

2	3
1	1

Check page 2

38–70 MIN

S4	1 and 3 in the top row; 1 and 1 in the bottom row
S5	A line across the middle, from the left edge to the right edge
S6	7 cubes

38–70 MIN

Choose your next page

Page	What it is
Think harder (page 3)	Hidden cubes, a fold and a hole, the other hole
Go further (page 4)	Fold twice, a bigger stack, build your own
Investigation (page 5)	Fold, mark and open
Check	Finish and check your own work

Everyone may choose. Not finished? Pages 3 to 5 are for home too.

Talk and explain

70–76 MIN

A CHILD IN ANOTHER CLASS SAYS

I count the cubes I can see in the picture.

I see 6 cubes.

So there are 6 cubes in the stack.

Do you agree? How could we check?

I think … because …

Back to the sneak-peek

70–76 MIN

Puzzle 1: 4 cubes. Puzzle 2: the 5th cube is hidden behind the front tower.

Exit ticket

76–82 MIN

On your own. No looking back.

E1 **Look at the cubes.**

There are ☐ cubes in the picture.



E2 **The top view shows how many cubes are in each stack.**

There are ☐ cubes in all.

2	1
2	1

E3 **A square is folded so the left edge meets the right edge.**

Where is the fold line? Circle one.

Check the exit ticket

76–82 MIN

E1	4 cubes
E2	6 cubes
E3	down the middle

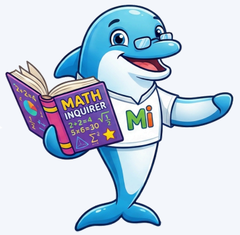
Look-back

82–85 MIN

Three minutes. There are no right answers. You can pass. Nobody reads it out.

Something I did well	write or draw one thing
The hardest part was …	☐ hidden cubes ☐ top views ☐ folding ☐ something else
What helped me	☐ my blocks ☐ top view numbers ☐ folding paper ☐ I was not stuck
Next time I will try	☐ hidden cubes ☐ top views ☐ holes ☐ asking a question
	☐ I am still stuck (my grown-up will photograph this page for my tutor)
	☐ I would like to pass

Take your time. You can pass.



85–90 MIN

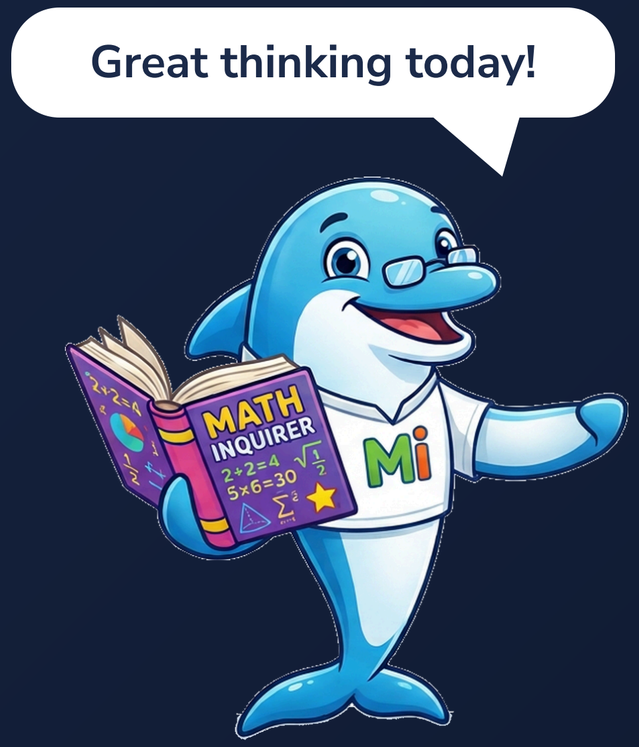
Well done, inquirers!

Show a grown-up a stack and its top view.

NEXT LESSON

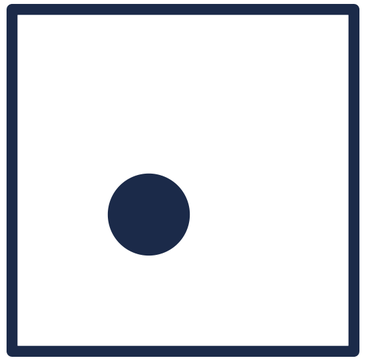
Spot It

Have ready: the 12 pair cards from your book, cut out by a grown-up.



Go further: fold twice

OPTIONAL



Folded in half, then in half again. One hole. How many holes when it is opened?

OPTIONAL

If we have time: pick one

Activity	What we do
Is it always true?	Does a bigger stack always have more cubes in the top view numbers?
Make a puzzle	Build a stack and draw its top view for your grown-up.
Teach it back	Show a toy why a picture can hide a cube.

Optional. Everyone may join.